# set_config

Command: `set_config`

## Description

Command issued to set the complete device configuration such as current region details, ethernet & wifi configuration, end point configuration, event and alert configuration

## Request

### Request Fields

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `command` | string | Yes | Specifies the operation being performed, in this case, setting the reader configuration. | `set_config` |
| `requestId` | string | Yes | A unique identifier for the request, allowing tracking and debugging of the operation. | `abc123` |
| `configData` | object | No | | |
| `configData.applyAfterReboot` | boolean | No | if set to true, configuration changes will be applied after reboot | |
| `configData.wifiConfig` | object | No | | |
| `configData.wifiConfig.interfaceName` | string | Yes | The name of the network interface for which the Wi-Fi configuration is being set. | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.wifiConfig.enableInterface` | boolean | Yes | Specifies whether the network interface should be enabled or disabled. | |
| `configData.wifiConfig.operation` | string (enum) | Yes | The operation to be performed on the Wi-Fi configuration. It can either create a new configuration or modify an existing one. | `CREATE` |
| `configData.wifiConfig.accessPoint` | object | Yes | | |
| `configData.wifiConfig.accessPoint.connect` | boolean | Yes | Indicates whether to connect to this profile. If true, disconnects the currently connected profile and connects to the specified one; if false, just saves the profile. | |
| `configData.wifiConfig.accessPoint.isPreferred` | boolean | Yes | Marks this profile as the preferred one to always connect to when available. | |
| `configData.wifiConfig.accessPoint.autoConn` | boolean | No | Reserved for backward compatibility. It will be removed in the future. | |
| `configData.wifiConfig.accessPoint.essid` | string | Yes | The ESSID (network name) of the access point. | `ZWireless` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.wifiConfig.accessPoint.enableSecurity` | boolean | Yes | Indicates whether security is enabled for this access point. | |
| `configData.wifiConfig.accessPoint.ipv4Configuration` | object | No | | |
| `configData.wifiConfig.accessPoint.ipv4Configuration.ipAddress` | string | No | The IPv4 address to assign to the interface. | `192.168.1.81` |
| `configData.wifiConfig.accessPoint.ipv4Configuration.subnetMask` | string | No | The IPv4 subnet mask. | `255.255.255.0` |
| `configData.wifiConfig.accessPoint.ipv4Configuration.gateway` | string | No | The IPv4 gateway address. | `192.168.1.144` |
| `configData.wifiConfig.accessPoint.ipv4Configuration.enableDhcp` | boolean | No | Specifies whether DHCP is enabled for automatic IP address assignment. | |
| `configData.wifiConfig.accessPoint.ipv4Configuration.dnsServer` | string | No | The IPv4 address of the DNS server. | `192.168.1.78` |
| `configData.wifiConfig.accessPoint.ipv4Configuration.domainName` | string | No | The domain name for the network. | `example.test.com` |
| `configData.wifiConfig.accessPoint.ipv6Configuration` | object | No | | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.wifiConfig.accessPoint.ipv6Configuration.ipAddress` | string | No | The IPv6 address to assign to the interface. | `2a02:1234:420a:100b::99` |
| `configData.wifiConfig.accessPoint.ipv6Configuration.prefix` | integer | No | The IPv6 address prefix length. | `65` |
| `configData.wifiConfig.accessPoint.ipv6Configuration.gateway` | string | No | The IPv6 gateway address. | `192.168.1.144` |
| `configData.wifiConfig.accessPoint.ipv6Configuration.enableAuto` | boolean | No | Specifies whether automatic IPv6 address assignment (e.g., SLAAC) is enabled. | |
| `configData.wifiConfig.accessPoint.ipv6Configuration.dnsServer` | string | No | The IPv6 address of the DNS server. | `2001:4860:4860::9999` |
| `configData.wifiConfig.accessPoint.ipv6Configuration.domainName` | string | No | The domain name for the IPv6 network. | `example.com` |
| `configData.wifiConfig.accessPoint.security` | object | No | | |
| `configData.wifiConfig.accessPoint.security.securityType` | string (enum) | Yes | The security type for the access point, specifying the authentication and encryption method. | `WPA2personal` |
| `configData.wifiConfig.accessPoint.security.securityDetails` | object | Yes | | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Personal` | object | No | | |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Personal.password` | string | Yes | The password for WPA2-Personal authentication. | `test@123` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Personal` | object | No | | |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Personal.password` | string | Yes | The password for WPA3-Personal authentication. | |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise` | object | No | WPA2-Enterprise security configuration with optional certificate parameters. | |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.authentication` | string (enum) | Yes | Authentication protocol for WPA2-Enterprise. | `peap` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.innerAuthentication` | string (enum) | No | Inner authentication method to be used within the outer authentication protocol. | `tls` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.identity` | string | No | The identity (username) for WPA2-Enterprise authentication. | `test` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.anonymousIdentity` | string | No | The anonymous identity for WPA2-Enterprise authentication. | `test` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.password` | string | No | The password for WPA2-Enterprise authentication. | `test@123` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.passphrase` | string | No | The passphrase for WPA2-Enterprise. | `test@123` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.certificate` | array | No | | `ap1wpa2e` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.protocol` | string (enum) | No | The encryption protocol for WPA2-Enterprise. | `WPA2_Enterprise_CCMP` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise` | object | No | WPA3-Enterprise security configuration with optional certificate parameters. | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.authentication` | string (enum) | Yes | Authentication protocol for WPA3-Enterprise. | `peap` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.innerAuthentication` | string (enum) | No | Inner authentication method to be used within the outer authentication protocol. | `tls` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.identity` | string | No | The identity (username) for WPA3-Enterprise authentication. | `test` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.anonymousIdentity` | string | No | The anonymous identity for WPA3-Enterprise authentication. | `test` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.password` | string | No | The password for WPA3-Enterprise authentication. | `test@123` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.passphrase` | string | No | The passphrase for WPA3-Enterprise. | `test@123` |
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.certificate` | array | No | | `ap1wpa2e` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.protocol` | string (enum) | No | The encryption protocol for WPA3-Enterprise. | `WPA3_Enterprise_CCMP` |
| `configData.wifiConfig.accessPoint.security.securityDetails.OWEPublic` | object | No | | |
| `configData.wifiConfig.accessPoint.security.securityDetails.OWEPublic.authentication` | string | Yes | Authentication protocol for OWE-Public. | |
| `configData.epConfig` | object | No | | |
| `configData.epConfig.operation` | string (enum) | Yes | Specifies the type of operation to perform, either adding or deleting a configuration. | `add` |
| `configData.epConfig.configuration` | object | Yes | | |
| `configData.epConfig.configuration.endpointName` | string | Yes | Identifies the name of the endpoint, which determines its purpose or function. | `mgmt` |
| `configData.epConfig.configuration.epType` | string (enum) | Yes | Specifies the type of the endpoint. | `MGMT` |
| `configData.epConfig.configuration.protocol` | string (enum) | Yes | Defines the communication protocol used by the endpoint. | `MQTT` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.epConfig.configuration.activate` | boolean | Yes | Indicates whether the endpoint is active or inactive. | |
| `configData.epConfig.configuration.url` | string | Yes | Specifies the URL associated with the endpoint. | `soti.example.com` |
| `configData.epConfig.configuration.verificationType` | string (enum) | Yes | Specifies the type of verification used for establishing secure connections. | `VERIFY_PEER` |
| `configData.epConfig.configuration.port` | integer | Yes | Specifies the port number used for communication. | `1883` |
| `configData.epConfig.configuration.qosCommon` | integer | Yes | Defines the Quality of Service (QoS) level for the endpoint. | `1` |
| `configData.epConfig.configuration.tenantId` | string | Yes | Identifies the tenantID associated with the configuration. | `ZEBRA` |
| `configData.epConfig.configuration.mqttParams` | object | No | | |
| `configData.epConfig.configuration.mqttParams.keepAlive` | integer | No | Specifies the keep-alive interval in seconds for MQTT connections. | `300` |
| `configData.epConfig.configuration.mqttParams.cleanSession` | boolean | No | Indicates whether to start a clean session for the MQTT client. | |
| `configData.epConfig.configuration.mqttParams.reconnectDelayMin` | integer | No | Specifies the minimum delay in seconds before reconnecting. | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.epConfig.configuration.mqttParams.reconnectDelayMax` | integer | No | Specifies the maximum delay in seconds before reconnecting. | |
| `configData.epConfig.configuration.mqttParams.clientId` | string | No | Specifies the client ID used for MQTT connections. | `zebra-rfd40-01` |
| `configData.epConfig.configuration.mqttParams.username` | string | No | The username for authentication in MQTT connections. | |
| `configData.epConfig.configuration.mqttParams.password` | string | No | The password for authentication in MQTT connections. | |
| `configData.epConfig.configuration.mqttParams.publishTopics` | array | No | | |
| `configData.epConfig.configuration.mqttParams.subscribeTopics` | array | No | | |
| `configData.epConfig.configuration.securityParams` | object | No | Defines the security parameters for the endpoint. | |
| `configData.epConfig.configuration.securityParams.format` | string (enum) | Yes | Specifies the format of the security certificates. Currently, only PEM is supported. | `PEM` |
| `configData.epConfig.configuration.securityParams.algorithm` | string (enum) | Yes | Specifies the algorithm used for encryption. | `RSA256` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.epConfig.configuration.securityParams.caCertificateFile` | string | Yes | Specifies the file containing the CA certificates. | `cloud-trusted-ca-certs.crt` |
| `configData.epConfig.configuration.securityParams.clientCert` | string | Yes | Specifies the file containing the client certificate. | `client_cert.pem` |
| `configData.epConfig.configuration.securityParams.clientKey` | string | Yes | Specifies the file containing the client private key. | `client_key.pem` |
| `configData.epConfig.configuration.eventConfiguration` | object | No | | |
| `configData.epConfig.configuration.eventConfiguration.antenna` | boolean | No | set to true to enable antenna events | |
| `configData.epConfig.configuration.eventConfiguration.terminalConnection` | boolean | No | set to true to enable terminal connection/disconnection events | |
| `configData.epConfig.configuration.eventConfiguration.firmwareUpdate` | boolean | No | set to true to enable fw update events | |
| `configData.epConfig.configuration.eventConfiguration.gpi` | boolean | No | set to true to enable gpi events | |
| `configData.epConfig.configuration.eventConfiguration.network` | boolean | No | set to true to enable network events | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.epConfig.configuration.eventConfiguration.exceptions` | boolean | No | set to true to enable exception events | |
| `configData.epConfig.configuration.eventConfiguration.ntp` | boolean | No | set to true to enable battery alerts | |
| `configData.epConfig.configuration.eventConfiguration.userApp` | boolean | No | set to true to enable user app events | |
| `configData.epConfig.configuration.eventConfiguration.heartbeat` | boolean | No | set to true to enable heart beat | |
| `configData.epConfig.configuration.eventConfiguration.power` | boolean | No | set to true to enable power source alerts | |
| `configData.epConfig.configuration.eventConfiguration.battery` | boolean | No | set to true to enable battery alerts | |
| `configData.epConfig.configuration.eventConfiguration.temperature` | boolean | No | set to true to enable temperature alerts | |
| `configData.epConfig.configuration.eventConfiguration.fileDownload` | boolean | No | set to true to enable file download alerts | |
| `configData.epConfig.configuration.eventConfiguration.cpuUsage` | boolean | No | set to true to enable cpu usage alerts | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.epConfig.configuration.eventConfiguration.flashUsage` | boolean | No | set to true to enable flash usage alerts | |
| `configData.epConfig.configuration.eventConfiguration.ramUsage` | boolean | No | set to true to enable ram usage alerts | |
| `configData.epConfig.configuration.eventConfiguration.heartbeatConfiguration` | object | No | | |
| `configData.epConfig.configuration.eventConfiguration.heartbeatConfiguration.interval` | integer | No | heart beat interval in seconds | `100` |
| `configData.epConfig.configuration.eventConfiguration.heartbeatConfiguration.inventoryStatus` | boolean | No | set to true to enable inventory status as part of heartbeat message | |
| `configData.epConfig.configuration.eventConfiguration.heartbeatConfiguration.batteryStatus` | boolean | No | set to true to enable battery status as part of heartbeat message | |
| `configData.epConfig.configuration.eventConfiguration.heartbeatConfiguration.userApps` | boolean | No | set to true to enable user app status as part of heartbeat message | |
| `configData.epConfig.configuration.eventConfiguration.cpuThreshold` | integer | No | cpu usage value(%) above which alert will be triggered | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `configData.epConfig.configuration.eventConfiguration.ramThreshold` | integer | No | ram usage value(%) above which alert will be triggered | |
| `configData.epConfig.configuration.eventConfiguration.flashThreshold` | integer | No | flash usage value(%) above which alert will be triggered | |
| `configData.epConfig.configuration.eventConfiguration.temperatureThreshold` | integer | No | temperature treshold value above which alert will be triggered | |

## Request Examples

### Create wifi profile

```
{
  "command": "set_config",
  "requestId": "209543",
  "configData": {
    "wifiConfig": {
      "operation": "CREATE",
      "accessPoint": {
        "essid": "TEST_5G",
        "connect": true,
        "autoConn": true,
        "enableSecurity": true,
        "isPreferred": false,
        "security": {
          "securityType": "WPA2Personal",
          "securityDetails": {
            "WPA2Personal": {
              "password": "nov211985"
            }
          }
        }
      }
    }
```

```
  }
}
```

### Modify wifi profile

```
{
  "command": "set_config",
  "requestId": "209543",
  "configData": {
    "wifiConfig": {
      "operation": "MODIFY",
      "accessPoint": {
        "essid": "TEST_5G",
        "connect": true,
        "autoConn": true,
        "enableSecurity": true,
        "isPreferred": true,
        "security": {
          "securityType": "WPA2Personal",
          "securityDetails": {
            "WPA2Personal": {
              "password": "nov211985"
            }
          }
        }
      }
    }
  }
}
```

### Add mdm endpoint

```
{
  "command": "set_config",
  "requestId": "209543",
  "configData": {
    "applyAfterReboot": true,
    "epConfig": {
      "operation": "add",
      "configuration": {
        "endpointName": "gmdm",
        "epType": "MDM",
        "mqttParams": {
          "keepAlive": 300
```

```
        }
      }
    }
  }
}
```

### Add data endpoint

```
{
  "command": "set_config",
  "requestId": "1233",
  "configData": {
    "epConfig": {
      "operation": "add",
      "configuration": {
        "endpointName": "dataEP",
        "epType": "DATA1",
        "protocol": "MQTT",
        "activate": true,
        "url": "17L11-GG6663-1",
        "verificationType": "NONE",
        "port": 1883,
        "qosCommon": 1,
        "tenantId": "ZEBRA",
        "mqttParams": {
          "keepAlive": 300,
          "cleanSession": true,
          "reconnectDelayMin": 50,
          "reconnectDelayMax": 500,
          "clientId": "zebra-rfd40-01",
          "username": "username",
          "password": "password"
        }
      }
    }
  }
}
```

### Add control endpoint

```
{
  "command": "set_config",
  "requestId": "1234",
  "configData": {
```

```
    "epConfig": {
      "operation": "add",
      "configuration": {
        "endpointName": "ctrlEP",
        "epType": "CTRL",
        "protocol": "MQTT",
        "activate": true,
        "url": "17L11-GG6663-1",
        "verificationType": "NONE",
        "port": 1883,
        "qosCommon": 1,
        "tenantId": "zebra",
        "mqttParams": {
          "keepAlive": 300,
          "cleanSession": true,
          "reconnectDelayMin": 50,
          "reconnectDelayMax": 500,
          "clientId": "zebra-rfd40-01",
          "username": "mqttuser",
          "password": "password",
          "publishTopics": [
            {
              "topic": "MDM/clients/resp",
              "qos": 1,
              "retain": false
            },
            {
              "topic": "MDM/clients/event",
              "qos": 1,
              "retain": false
            },
            {
              "topic": "MDM/clients/rfid",
              "qos": 0,
              "retain": true
            }
          ],
          "subscribeTopics": [
            {
              "topic": "MDM/clients/cmnd",
              "qos": 0,
              "retain": false
            }
          ]
        }
```

```
      }
    }
  }
}
```

### Complete config setup

```
{
  "command": "set_config",
  "requestId": "5678",
  "configData": {
    "wifiConfig": {
      "operation": "CREATE",
      "accessPoint": {
        "essid": "CORP_WIFI",
        "connect": true,
        "autoConn": false,
        "enableSecurity": true,
        "isPreferred": true,
        "security": {
          "securityType": "WPA2Personal",
          "securityDetails": {
            "WPA2Personal": {
              "password": "corp_pass_2024"
            }
          }
        }
      }
    },
    "epConfig": {
      "operation": "add",
      "configuration": {
        "endpointName": "mgmtEp",
        "epType": "MGMT",
        "protocol": "MQTT",
        "activate": true,
        "url": "iot-broker.company.com",
        "verificationType": "NONE",
        "port": 8883,
        "qosCommon": 1,
        "tenantId": "ZEBRA",
        "mqttParams": {
          "keepAlive": 60,
          "cleanSession": true,
          "reconnectDelayMin": 10,
```

```
          "reconnectDelayMax": 120,
          "clientId": "iot",
          "username": "iotuser",
          "password": "iotpassword",
          "publishTopics": [
            {
              "topic": "IOT/data/resp",
              "qos": 1,
              "retain": false
            }
          ],
          "subscribeTopics": [
            {
              "topic": "IOT/data/cmnd",
              "qos": 1,
              "retain": false
            }
          ]
        }
      }
    }
  }
}
```

## Response

### Response Fields

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `command` | string | Yes | | `set_config` |
| `requestId` | string | Yes | | `abc123` |
| `apiVersion` | string (enum) | Yes | | `V1.1` |
| `cfgResponse` | object | No | | |
| `cfgResponse.ethConfig` | object | No | | |
| `cfgResponse.ethConfig.interfaceName` | string | Yes | Name of the network interface (e.g., eth0). | `eth0` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.ethConfig.enableInterface` | boolean | Yes | Indicates whether the network interface is enabled. | |
| `cfgResponse.ethConfig.lldp` | boolean | Yes | Specifies if Link Layer Discovery Protocol (LLDP) is enabled. | |
| `cfgResponse.ethConfig.ipv4Configuration` | object | No | Configuration settings for IPv4. | |
| `cfgResponse.ethConfig.ipv4Configuration.ipAddress` | string | No | The IPv4 address assigned to the interface. | `192.168.1.81` |
| `cfgResponse.ethConfig.ipv4Configuration.subnetMask` | string | No | The subnet mask associated with the IPv4 address. | `255.255.255.0` |
| `cfgResponse.ethConfig.ipv4Configuration.gateway` | string | No | The default gateway for the IPv4 network. | `192.168.1.144` |
| `cfgResponse.ethConfig.ipv4Configuration.enableDhcp` | boolean | No | Indicates whether DHCP is enabled for IPv4. | |
| `cfgResponse.ethConfig.ipv4Configuration.dnsServer` | string | No | The IPv4 address of the DNS server. | `192.168.1.78` |
| `cfgResponse.ethConfig.ipv4Configuration.domainName` | string | No | The domain name associated with the network. | `example.test.com` |
| `cfgResponse.ethConfig.ipv6Configuration` | object | No | Configuration settings for IPv6. | |
| `cfgResponse.ethConfig.ipv6Configuration.ipAddress` | string | No | The IPv6 address assigned to the interface. | `2a02:1234:420a:100b::99` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.ethConfig.ipv6Configuration.prefix` | integer | No | The prefix length for the IPv6 address. | `64` |
| `cfgResponse.ethConfig.ipv6Configuration.gateway` | string | No | The default gateway for the IPv6 network. | `fe80::1643:8376:b32:4a16` |
| `cfgResponse.ethConfig.ipv6Configuration.enableAuto` | boolean | No | Indicates whether auto-configuration is enabled for IPv6. | |
| `cfgResponse.ethConfig.ipv6Configuration.dnsServer` | string | No | The IPv6 address of the DNS server. | `2001:4860:4860::9999` |
| `cfgResponse.ethConfig.ipv6Configuration.domainName` | string | No | The domain name associated with the IPv6 network. | `example.com` |
| `cfgResponse.ethConfig.security` | object | No | Security configuration for the network interface. | |
| `cfgResponse.ethConfig.security.enableSecurity` | boolean | No | Indicates if security is enabled. If enabled, 802.X parameters must be provided. | |
| `cfgResponse.ethConfig.security.EAP802_1X` | object | No | Extensible Authentication Protocol (EAP) 802.1X settings. | |
| `cfgResponse.ethConfig.security.EAP802_1X.authentication` | string (enum) | No | The authentication method used for EAP 802.1X. | `PEAP` |
| `cfgResponse.ethConfig.security.EAP802_1X.innerAuthentication` | string (enum) | No | The inner authentication method used within the EAP tunnel. | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.ethConfig.security.EAP802_1X.certificate` | string | No | The certificate used for EAP authentication. | `ethCert` |
| `cfgResponse.wifiConfigResp` | object | No | | |
| `cfgResponse.wifiConfigResp.interfaceName` | string | Yes | The name of the network interface for which the Wi-Fi configuration is being set. | |
| `cfgResponse.wifiConfigResp.enableInterface` | boolean | Yes | Specifies whether the network interface should be enabled or disabled. | |
| `cfgResponse.wifiConfigResp.operation` | string (enum) | Yes | The operation to be performed on the Wi-Fi configuration. It can either create a new configuration or modify an existing one. | `CREATE` |
| `cfgResponse.wifiConfigResp.accessPoint` | object | Yes | | |
| `cfgResponse.wifiConfigResp.accessPoint.connect` | boolean | Yes | Indicates whether to connect to this profile. If true, disconnects the currently connected profile and connects to the specified one; if false, just saves the profile. | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.wifiConfigResp.accessPoint.isPreferred` | boolean | Yes | Marks this profile as the preferred one to always connect to when available. | |
| `cfgResponse.wifiConfigResp.accessPoint.autoConn` | boolean | No | Reserved for backward compatibility. It will be removed in the future. | |
| `cfgResponse.wifiConfigResp.accessPoint.essid` | string | Yes | The ESSID (network name) of the access point. | `ZWireless` |
| `cfgResponse.wifiConfigResp.accessPoint.enableSecurity` | boolean | Yes | Indicates whether security is enabled for this access point. | |
| `cfgResponse.wifiConfigResp.accessPoint.ipv4Configuration` | object | No | | |
| `cfgResponse.wifiConfigResp.accessPoint.ipv4Configuration.ipAddress` | string | No | The IPv4 address to assign to the interface. | `192.168.1.81` |
| `cfgResponse.wifiConfigResp.accessPoint.ipv4Configuration.subnetMask` | string | No | The IPv4 subnet mask. | `255.255.255.0` |
| `cfgResponse.wifiConfigResp.accessPoint.ipv4Configuration.gateway` | string | No | The IPv4 gateway address. | `192.168.1.144` |
| `cfgResponse.wifiConfigResp.accessPoint.ipv4Configuration.enableDhcp` | boolean | No | Specifies whether DHCP is enabled for automatic IP address assignment. | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.wifiConfigResp.accessPoint.ipv4Configuration.dnsServer` | string | No | The IPv4 address of the DNS server. | `192.168.1.78` |
| `cfgResponse.wifiConfigResp.accessPoint.ipv4Configuration.domainName` | string | No | The domain name for the network. | `example.test.com` |
| `cfgResponse.wifiConfigResp.accessPoint.ipv6Configuration` | object | No | | |
| `cfgResponse.wifiConfigResp.accessPoint.ipv6Configuration.ipAddress` | string | No | The IPv6 address to assign to the interface. | `2a02:1234:420a:100b::99` |
| `cfgResponse.wifiConfigResp.accessPoint.ipv6Configuration.prefix` | integer | No | The IPv6 address prefix length. | `65` |
| `cfgResponse.wifiConfigResp.accessPoint.ipv6Configuration.gateway` | string | No | The IPv6 gateway address. | `192.168.1.144` |
| `cfgResponse.wifiConfigResp.accessPoint.ipv6Configuration.enableAuto` | boolean | No | Specifies whether automatic IPv6 address assignment (e.g., SLAAC) is enabled. | |
| `cfgResponse.wifiConfigResp.accessPoint.ipv6Configuration.dnsServer` | string | No | The IPv6 address of the DNS server. | `2001:4860:4860::9999` |
| `cfgResponse.wifiConfigResp.accessPoint.ipv6Configuration.domainName` | string | No | The domain name for the IPv6 network. | `example.com` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.wifiConfigResp.accessPoint.security` | object | No | | |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityType` | string (enum) | Yes | The security type for the access point, specifying the authentication and encryption method. | `WPA2personal` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails` | object | Yes | | |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Personal` | object | No | | |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Personal.password` | string | Yes | The password for WPA2-Personal authentication. | `test@123` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Personal` | object | No | | |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Personal.password` | string | Yes | The password for WPA3-Personal authentication. | |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Enterprise` | object | No | WPA2-Enterprise security configuration with optional certificate parameters. | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Enterprise.authentication` | string (enum) | Yes | Authentication protocol for WPA2-Enterprise. | `peap` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Enterprise.innerAuthentication` | string (enum) | No | Inner authentication method to be used within the outer authentication protocol. | `tls` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Enterprise.identity` | string | No | The identity (username) for WPA2-Enterprise authentication. | `test` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Enterprise.anonymousIdentity` | string | No | The anonymous identity for WPA2-Enterprise authentication. | `test` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Enterprise.password` | string | No | The password for WPA2-Enterprise authentication. | `test@123` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Enterprise.passphrase` | string | No | The passphrase for WPA2-Enterprise. | `test@123` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Enterprise.certificate` | array | No | | `ap1wpa2e` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA2Enterprise.protocol` | string (enum) | No | The encryption protocol for WPA2-Enterprise. | `WPA2_Enterprise_CCMP` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Enterprise` | object | No | WPA3-Enterprise security configuration with optional certificate parameters. | |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Enterprise.authentication` | string (enum) | Yes | Authentication protocol for WPA3-Enterprise. | `peap` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Enterprise.innerAuthentication` | string (enum) | No | Inner authentication method to be used within the outer authentication protocol. | `tls` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Enterprise.identity` | string | No | The identity (username) for WPA3-Enterprise authentication. | `test` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Enterprise.anonymousIdentity` | string | No | The anonymous identity for WPA3-Enterprise authentication. | `test` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Enterprise.password` | string | No | The password for WPA3-Enterprise authentication. | `test@123` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Enterprise.passphrase` | string | No | The passphrase for WPA3-Enterprise. | `test@123` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Enterprise.certificate` | array | No | | `ap1wpa2e` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.WPA3Enterprise.protocol` | string (enum) | No | The encryption protocol for WPA3-Enterprise. | `WPA3_Enterprise_CCMP` |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.OWEPublic` | object | No | | |
| `cfgResponse.wifiConfigResp.accessPoint.security.securityDetails.OWEPublic.authentication` | string | Yes | Authentication protocol for OWE-Public. | |
| `cfgResponse.epConfig` | object | No | | |
| `cfgResponse.epConfig.operation` | string (enum) | Yes | Specifies the type of operation to perform, either adding or deleting a configuration. | `add` |
| `cfgResponse.epConfig.configuration` | object | Yes | | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.epConfig.configuration.endpointName` | string | Yes | Identifies the name of the endpoint, which determines its purpose or function. | `mgmt` |
| `cfgResponse.epConfig.configuration.epType` | string (enum) | Yes | Specifies the type of the endpoint. | `MGMT` |
| `cfgResponse.epConfig.configuration.protocol` | string (enum) | Yes | Defines the communication protocol used by the endpoint. | `MQTT` |
| `cfgResponse.epConfig.configuration.activate` | boolean | Yes | Indicates whether the endpoint is active or inactive. | |
| `cfgResponse.epConfig.configuration.url` | string | Yes | Specifies the URL associated with the endpoint. | `soti.example.com` |
| `cfgResponse.epConfig.configuration.verificationType` | string (enum) | Yes | Specifies the type of verification used for establishing secure connections. | `VERIFY_PEER` |
| `cfgResponse.epConfig.configuration.port` | integer | Yes | Specifies the port number used for communication. | `1883` |
| `cfgResponse.epConfig.configuration.qosCommon` | integer | Yes | Defines the Quality of Service (QoS) level for the endpoint. | `1` |
| `cfgResponse.epConfig.configuration.tenantId` | string | Yes | Identifies the tenantID associated with the configuration. | `ZEBRA` |
| `cfgResponse.epConfig.configuration.mqttParams` | object | No | | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.epConfig.configuration.mqttParams.keepAlive` | integer | No | Specifies the keep-alive interval in seconds for MQTT connections. | `300` |
| `cfgResponse.epConfig.configuration.mqttParams.cleanSession` | boolean | No | Indicates whether to start a clean session for the MQTT client. | |
| `cfgResponse.epConfig.configuration.mqttParams.reconnectDelayMin` | integer | No | Specifies the minimum delay in seconds before reconnecting. | |
| `cfgResponse.epConfig.configuration.mqttParams.reconnectDelayMax` | integer | No | Specifies the maximum delay in seconds before reconnecting. | |
| `cfgResponse.epConfig.configuration.mqttParams.clientId` | string | No | Specifies the client ID used for MQTT connections. | `zebra-rfd40-01` |
| `cfgResponse.epConfig.configuration.mqttParams.username` | string | No | The username for authentication in MQTT connections. | |
| `cfgResponse.epConfig.configuration.mqttParams.password` | string | No | The password for authentication in MQTT connections. | |
| `cfgResponse.epConfig.configuration.mqttParams.publishTopics` | array | No | | |
| `cfgResponse.epConfig.configuration.mqttParams.subscribeTopics` | array | No | | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.epConfig.configuration.securityParams` | object | No | Defines the security parameters for the endpoint. | |
| `cfgResponse.epConfig.configuration.securityParams.format` | string (enum) | Yes | Specifies the format of the security certificates. Currently, only PEM is supported. | `PEM` |
| `cfgResponse.epConfig.configuration.securityParams.algorithm` | string (enum) | Yes | Specifies the algorithm used for encryption. | `RSA256` |
| `cfgResponse.epConfig.configuration.securityParams.caCertificateFile` | string | Yes | Specifies the file containing the CA certificates. | `cloud-trusted-ca-certs.crt` |
| `cfgResponse.epConfig.configuration.securityParams.clientCert` | string | Yes | Specifies the file containing the client certificate. | `client_cert.pem` |
| `cfgResponse.epConfig.configuration.securityParams.clientKey` | string | Yes | Specifies the file containing the client private key. | `client_key.pem` |
| `cfgResponse.epConfig.configuration.eventConfiguration` | object | No | | |
| `cfgResponse.epConfig.configuration.eventConfiguration.antenna` | boolean | No | set to true to enable antenna events | |
| `cfgResponse.epConfig.configuration.eventConfiguration.terminalConnection` | boolean | No | set to true to enable terminal connection/disconnection events | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.epConfig.configuration.eventConfiguration.firmwareUpdate` | boolean | No | set to true to enable fw update events | |
| `cfgResponse.epConfig.configuration.eventConfiguration.gpi` | boolean | No | set to true to enable gpi events | |
| `cfgResponse.epConfig.configuration.eventConfiguration.network` | boolean | No | set to true to enable network events | |
| `cfgResponse.epConfig.configuration.eventConfiguration.exceptions` | boolean | No | set to true to enable exception events | |
| `cfgResponse.epConfig.configuration.eventConfiguration.ntp` | boolean | No | set to true to enable battery alerts | |
| `cfgResponse.epConfig.configuration.eventConfiguration.userApp` | boolean | No | set to true to enable user app events | |
| `cfgResponse.epConfig.configuration.eventConfiguration.heartbeat` | boolean | No | set to true to enable heart beat | |
| `cfgResponse.epConfig.configuration.eventConfiguration.power` | boolean | No | set to true to enable power source alerts | |
| `cfgResponse.epConfig.configuration.eventConfiguration.battery` | boolean | No | set to true to enable battery alerts | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.epConfig.configuration.eventConfiguration.temperature` | boolean | No | set to true to enable temperature alerts | |
| `cfgResponse.epConfig.configuration.eventConfiguration.fileDownload` | boolean | No | set to true to enable file download alerts | |
| `cfgResponse.epConfig.configuration.eventConfiguration.cpuUsage` | boolean | No | set to true to enable cpu usage alerts | |
| `cfgResponse.epConfig.configuration.eventConfiguration.flashUsage` | boolean | No | set to true to enable flash usage alerts | |
| `cfgResponse.epConfig.configuration.eventConfiguration.ramUsage` | boolean | No | set to true to enable ram usage alerts | |
| `cfgResponse.epConfig.configuration.eventConfiguration.heartbeatConfiguration` | object | No | | |
| `cfgResponse.epConfig.configuration.eventConfiguration.heartbeatConfiguration.interval` | integer | No | heart beat interval in seconds | `100` |
| `cfgResponse.epConfig.configuration.eventConfiguration.heartbeatConfiguration.inventoryStatus` | boolean | No | set to true to enable inventory status as part of heartbeat message | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `cfgResponse.epConfig.configuration.eventConfiguration.heartbeatConfiguration.batteryStatus` | boolean | No | set to true to enable battery status as part of heartbeat message | |
| `cfgResponse.epConfig.configuration.eventConfiguration.heartbeatConfiguration.userApps` | boolean | No | set to true to enable user app status as part of heartbeat message | |
| `cfgResponse.epConfig.configuration.eventConfiguration.cpuThreshold` | integer | No | cpu usage value(%) above which alert will be triggered | |
| `cfgResponse.epConfig.configuration.eventConfiguration.ramThreshold` | integer | No | ram usage value(%) above which alert will be triggered | |
| `cfgResponse.epConfig.configuration.eventConfiguration.flashThreshold` | integer | No | flash usage value(%) above which alert will be triggered | |
| `cfgResponse.epConfig.configuration.eventConfiguration.temperatureThreshold` | integer | No | temperature treshold value above which alert will be triggered | |
| `cfgResponse.wifiProfiles` | object | No | | |
| `cfgResponse.wifiProfiles.wifiConfig` | array | No | | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `response` | object | Yes | Status of the command execution — check `code` to confirm success (0) or identify the error. | |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| `response.code` | integer | Yes | Response code indicating success or failure.<br>• **0** — Success<br>• **1** — Command payload accepted<br>• **2** — Invalid payload<br>• **3** — Not able to retrieve information<br>• **4** — Firmware update in progress<br>• **5** — Can't reboot, inventory in progress<br>• **6** — Region is not configured<br>• **7** — Interface is not available<br>• **8** — Insufficient flash size<br>• **9** — File not found<br>• **10** — Configuration already exists<br>• **11** — Inventory in progress<br>• **12** — No radio operation in progress<br>• **13** — Firmware update failed<br>• **14** — Battery too low for firmware update<br>• **15** — WIFI: SSID not found | `0` |

| FIELD | TYPE | REQUIRED | DESCRIPTION | EXAMPLE |
|---|---|---|---|---|
| | | | • **16** — WIFI: Cannot delete active SSID<br>• **17** — WIFI: SSID missing<br>• **18** — WIFI: SSID already exists<br>• **19** — WIFI: SSID count overflow<br>• **20** — WiFi is not supported<br>• **21** — Certificate not found<br>• **22** — Advanced configuration not set<br>• **23** — Invalid enum value<br>• **24** — Max 32 prefilters limit exceeded<br>• **25** — Max 3 publish topics exceeded<br>• **26** — Max 1 subscribe topic exceeded<br>• **27** — Invalid tenant ID length<br>• **28** — Tag match pattern length exceeded | |
| `response.description` | string | Yes | Human-readable description of the response code (e.g. "Success", "Invalid payload"). | `Success` |

## Response Examples

### example1

```
{
  "command": "set_config",
  "requestId": "209543",
  "apiVersion": "V1.1",
  "cfgResponse": {
    "wifiConfig": {
      "operation": "CREATE",
      "accessPoint": {
        "essid": "TEST_5G",
        "connect": true,
        "autoConn": true,
        "enableSecurity": true,
        "isPreferred": false,
        "security": {
          "securityType": "WPA2Personal",
          "securityDetails": {
            "WPA2Personal": {
              "password": "XXXXXXXX"
            }
          }
        }
      }
    }
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```

### example2

```
{
  "command": "set_config",
  "requestId": "209543",
  "apiVersion": "V1.1",
  "cfgResponse": {
    "wifiConfig": {
      "operation": "MODIFY",
      "accessPoint": {
        "essid": "TEST_5G",
```

```
        "connect": true,
        "autoConn": true,
        "enableSecurity": true,
        "isPreferred": true,
        "security": {
          "securityType": "WPA2Personal",
          "securityDetails": {
            "WPA2Personal": {
              "password": "XXXXXXXX"
            }
          }
        }
      }
    }
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```

### example3

```
{
  "command": "set_config",
  "requestId": "209543",
  "apiVersion": "V1.1",
  "cfgResponse": {
    "applyAfterReboot": true,
    "epConfig": {
      "operation": "add",
      "configuration": {
        "endpointName": "gmdm",
        "epType": "MDM",
        "mqttParams": {
          "keepAlive": 300
        }
      }
    }
  },
  "response": {
    "code": 0,
    "description": "Success"
```

```
  }
}
```

### example4

```
{
  "command": "set_config",
  "requestId": "1233",
  "apiVersion": "V1.1",
  "cfgResponse": {
    "epConfig": {
      "operation": "add",
      "configuration": {
        "endpointName": "dataEP",
        "epType": "DATA1",
        "protocol": "MQTT",
        "activate": true,
        "url": "17L11-GG6663-1",
        "verificationType": "NONE",
        "port": 1883,
        "qosCommon": 1,
        "tenantId": "ZEBRA",
        "mqttParams": {
          "keepAlive": 300,
          "cleanSession": true,
          "reconnectDelayMin": 50,
          "reconnectDelayMax": 500,
          "clientId": "zebra-rfd40-01",
          "username": "XXXXXXXX",
          "password": "XXXXXXXX"
        }
      }
    }
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```

### example5

```
{
  "command": "set_config",
```

```
  "requestId": "1234",
  "apiVersion": "V1.1",
  "cfgResponse": {
    "epConfig": {
      "operation": "add",
      "configuration": {
        "endpointName": "ctrlEP",
        "epType": "CTRL",
        "protocol": "MQTT",
        "activate": true,
        "url": "17L11-GG6663-1",
        "verificationType": "NONE",
        "port": 1883,
        "qosCommon": 1,
        "tenantId": "zebra",
        "mqttParams": {
          "keepAlive": 300,
          "cleanSession": true,
          "reconnectDelayMin": 50,
          "reconnectDelayMax": 500,
          "clientId": "zebra-rfd40-01",
          "username": "XXXXXXXX",
          "password": "XXXXXXXX",
          "publishTopics": [
            {
              "topic": "MGMT/clients/cmnd",
              "qos": 0,
              "retain": false
            }
          ],
          "subscribeTopics": [
            {
              "topic": "MGMT/clients/resp",
              "qos": 1,
              "retain": false
            },
            {
              "topic": "MGMT/clients/event",
              "qos": 1,
              "retain": false
            },
            {
              "topic": "MGMT/clients/rfid",
              "qos": 2,
              "retain": true
```

```
            }
          ]
        }
      }
    }
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```

### example6

```
{
  "command": "set_config",
  "requestId": "5678",
  "apiVersion": "V1.1",
  "cfgResponse": {
    "wifiConfig": {
      "operation": "CREATE",
      "accessPoint": {
        "essid": "CORP_WIFI",
        "connect": true,
        "autoConn": false,
        "enableSecurity": true,
        "isPreferred": true,
        "security": {
          "securityType": "WPA2Personal",
          "securityDetails": {
            "WPA2Personal": {
              "password": "XXXXXXXX"
            }
          }
        }
      }
    },
    "epConfig": {
      "operation": "add",
      "configuration": {
        "endpointName": "mgmtEp",
        "epType": "MGMT",
        "protocol": "MQTT",
        "activate": true,
        "url": "iot-broker.company.com",
```

```
        "verificationType": "NONE",
        "port": 8883,
        "qosCommon": 1,
        "tenantId": "ZEBRA",
        "mqttParams": {
          "keepAlive": 60,
          "cleanSession": true,
          "reconnectDelayMin": 10,
          "reconnectDelayMax": 120,
          "clientId": "iot",
          "username": "XXXXXXXX",
          "password": "XXXXXXXX",
          "publishTopics": [
            {
              "topic": "IOT/data/cmnd",
              "qos": 1,
              "retain": false
            }
          ],
          "subscribeTopics": [
            {
              "topic": "IOT/data/resp",
              "qos": 1,
              "retain": false
            }
          ]
        }
      }
    }
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```